8. Stochastic POS Tagging Using Probabilities

Aim: To implement a simple stochastic POS tagger using probabilities to assign tags to words.

Procedure:

1. Define words and their possible POS tags.
2. Assign probabilities to each tag.
3. Select the tag with the highest probability.
4. Display each word with its predicted tag.

Code:

```
tag_prob = {
    "book": {"NN": 0.8, "VB": 0.2},
    "runs": {"VB": 0.9, "NN": 0.1},
    "good": {"JJ": 0.95, "NN": 0.05}
}
words = ["book", "runs", "good"]
for word in words:
    tag = max(tag_prob[word], key=tag_prob[word].get)
    print(word, ":", tag)
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/8.py"
book : NN
runs : VB
good : JJ
```

Result: A simple probabilistic model successfully assigned POS tags based on predefined probabilities.